Evaluating Transformer-Based and Traditional Machine Learning Models for Sarcasm-Masked Toxicity Detection in Social Media Comments on American Tariffs

GGI Gamanayake [1#], MGKS Gamage [2], and DD Weerasuriya[3]

[1]*Department of Languages, General Sir John Kotelawala Defence University, Sri Lanka*
[2]*Department of Languages, General Sir John Kotelawala Defence University, Sri Lanka*
[3] *Department of Languages, General Sir John Kotelawala Defence University, Sri Lanka*
[#] <41-adc-0017@kdu.ac.lk>

Abstract**—** *Detection of toxicity in online conversations is important now more than ever, as harmful content is spread through social media. Sarcasm can be, however, a concealment of toxicity and therefore the detection systems can be less efficient by automation. Although some authors have investigated toxicity detection as well as sarcasm detection separately, there is little literature that rigorously compares the performance of transformer-based and traditional models in detecting sarcasm-masked toxic posts, especially in the politically sensitive issue of American tariffs. This paper compares DistilBERT, which is a transformer model, against other classic machine learning algorithms, such as the Support Vector Machine (SVM), the XGBoost, the Logistic Regression, and the Random Forest. Models are evaluated using a labeled set of social media comments about American tariffs based on Macro F1-score, sarcastic-toxic F1-score, precision, recall, accuracy, and confusion Matrises. Findings show that DistilBERT significantly outperforms classical models with a Macro F1-score of 0.65 and sarcastic-toxic F1-score of 0.67 and SVM and other baselines were impossible to perform on the sarcasm-masked toxicity (F1 ≈ 0.00). Results highlight the relevance of contextual language models in the moderation of political speech.*

Keywords**— Toxicity detection, Sarcasm detection, DistilBERT, Transformers, Social media, Tariffs, Machine learning**

I. INTRODUCTION

A rapid increase of social media services has changed the manner in which people communicate worldwide as people are able to express their views, conduct arguments and share their ideas on an international level [1], [15], [19], [25]. "Although this interconnectivity promotes democratization, on the other hand, it has contributed to the spread of negative content in the form of hate speech, harassment, threats, and abusive language" [2], [3], [15], [27],[31]. The need to identify and manage the toxic behaviour on the internet has consequently led to the need to safeguard constructive conversations and safeguard users against psychological damages [6], [24], [26].

The traditional methods of detection of toxicity have been based on extensive application of key word filtering and rule-based mechanisms [4], [25], [39]. Though these methods are able to identify explicit abuse, they cannot always work with the toxicity hidden behind sarcasm [5], [7], [23].

Sarcasm is another important issue, because it usually switches to negativity and communicates in an indirect manner, and thus the latest toxicity classifiers that are state-of-the-art will be seen as such by misinterpretation [7], [16], [28]. Research indicates that sarcasm does not only make it difficult to recognize toxicity, but it also has affecting effects on the feelings and even the extent of offensive material [24], [29], [44]. In order to deal with such complexity, scientists have resorted more and more to machine learning and natural language processing (NLP) models. Handcrafted features that were examined early on included n-grams, sentiment polarity shifts, and contextual incongruity [9], [31], [32], [48], but such approaches were found to be weak in revealing the subtleties of how sarcasm is expressed [30], [34]. Deep learning and transformer architectures turned everything upside down and can offer an embedding that is contextual and attention networks that bring significant accuracy to detection [11], [12],[13], [14]. "BERT, RoBERTa, and DistilBERT are models that have demonstrated state-of-the-art results in sentiment analysis, hate speech classification, and sarcasm recognition"[8], [12], [21], [22], [41],[47].

Although these advances are made, the issue of sarcasm as a phenomenon of toxicity has not been studied [10], [17], [18], [38]. Sarcasm in political and economic discourse like the case of American tariffs is a common rhetorical device, and it is challenging to figure out which discourse is harmful [20], [23], [36]. The proposed research seeks to address this gap by comparing in-depth transformer-based models, especially the DistilBERT, with the classical machine learning models, which include Support Vector Machine (SVM), Logistic Regression, Random Forest, and XGBoost to detect sarcasm-camouflaged toxicity [1].

Through the body of models, we expect to identify the best approaches that can address the intricate relationship of sarcasm and toxic intent in online comments.

This paper is expected to help the creation of more context-sensitive moderation tools that are able to effectively identify toxicity with the help of sarcasm, and, at the same time, maintain the variety and the multifacetedness of online communication. Old methods of toxicity detection, however, have failed to detect subtle types of harmful language, especially when the meaning is inverted or hidden behind the use of sarcasm. This study will compare the transformer-based models (i.e., DistilBERT) with classic machine learning methods (i.e., Support Vector Machines, Random Forests, Logistic Regression, and XGBoost) in the process of dealing with sarcasm-veiled toxicity, therefore, to understand which of these strategies has the least drawbacks and the most significant benefits [28].One of the focuses of the study is the ability to utilize the presence of contextual embeddings and attention mechanisms of transformer-based models. Embeddings in context enable the model to learn the sense of words within the context of the surrounding text, which is essential to identify when a given statement that appears positive is made with a bad or toxic purpose [26]. The attention mechanisms also increase the accuracy of the detection by allowing the model to prioritize the contribution of various words or phrases in a sentence to detect subtle indicators of sarcasm, which the traditional models fail to capture [14]. This research shows the practical benefit of the modern NLP methods as compared to classical methods by giving empirical evidence of how this mechanism enhances the performance of the model [6].Furthermore, the proposed research will be able to provide a foundation that makes it possible to apply the findings in the context of social media where the judicious use of moderation may increase user security without simplifying the online debate to an acceptable degree [41]. This is possible by enhancing the possibility of identifying sarcastic-toxic statements to curb the dissemination of harmful messages without damaging the expressive richness of online dialogs, or, more specifically, political, social, and economic discourses. Finally, the study will strive to enlighten the development of intelligent moderation systems that trade safety.

Lastly, the study aims at filling the gap between traditional machine learning algorithms and the existing transformer-based techniques, which can be useful in the creation of intelligent moderation systems [36]. Through a comparative analysis of both traditional algorithms (such as SVM, Random Forest and XGBoost) and more advanced transformer models (such as DistilBERT), the study will be able to highlight the most efficient methods at detecting toxicity, particularly when the technique was used to conceal it in a sarcastic manner [14]. Those learned can be applied to the creation of moderation systems that can treat the nuances and intricacy of online communication, as well as detecting dangerous content with high precision. The implementation of such systems can be done to curb the adverse exchange between political, social, and economic discourses in the social media environment and curb the proliferation of abusive or disruptive information [2], [3], [7], [15], [45],[46].

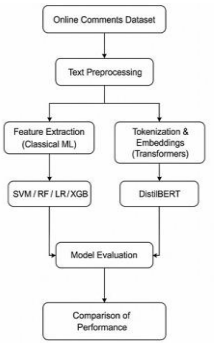


Figure 1. Conceptual Framework for Sarcasm-Masked Toxicity Detection

II. METHODOLOGY

A. Conceptual Framework

The proposed research will compare the transformer-based models and classical machine learning algorithms to identify sarcasm-masked toxicity in online comments in a systematic way [3]. The framework combines the old-fashioned feature-driven and the modern context-driven techniques in comparison of their relative performance in the process of capturing subtle toxic expressions. The workflow starts by processing and collecting data, text cleaning, tokenizing, and dealing with class imbalance. In the case of classical machine learning models, n-gram, sentiment polarity, and contextual incongruity are extracted as the features and train algorithms, including Support vector machine (SVM), random forest, logistic regression, and XGBoost. In the transformer-based models, e.g. DistilBERT, the tokenized text is mapped into the contextual embeddings and attention mechanisms are used to focus on the subtle trends of sarcasm and toxicity [13], [1], [17], [9]. After the model has been trained, accuracy, precision, recall, and F1-score are calculated as measures of the execution when comparing the suitability of each method [30]. With such a dual methodology, one can obtain the strengths and weaknesses of both classical and modern model and get the idea of which methods are the most appropriate in a sarcasm-sensitive toxicity detector

[32]. Figure 1 Offers a systematic framework on the method of systematically evaluating the ability of various models to identify sarcasm Networked torture to eventually inform the design of context-sensitive moderation systems in social media.

B. Dataset Description

The research data analyzed in this study was obtained from TikTok, Instagram and YouTube comments about American tariff barriers. Finally, 297 comments were collected, consisting of 148 toxic and 149 non-toxic comments. Additionally, 140 comments were found to be sarcastic and 156 were not sarcastic. All comments were also categorized based on sentiment polarity – positive, negative and neutral.

To ensure consistency, some preprocessing methods like removing URLs, mentions, unnecessary symbols and emojis were performed. The annotations were done manually [25][33] and the toxicity and sentiment labels were set according to the language used and the tone [19][27]. Three independent annotators agreed on the sarcasm labels [48][49]. This process guaranteed an accurate and consistent dataset for model training and testing [6][36].

C. Preprocessing

The data set was preprocessed using several methods to ensure that it was clean and consistent enough to model [14][31]. First, URLs, user mentions, emojis, repeated characters and irrelevant punctuation were removed to lower noise and enhance the model performance [16][34][19][42]. Next, the words were separated from the text to use word-level processing [14][50]. Low semantic value stopwords were eliminated to enhance the efficiency [14][31]. Additionally, stemming or lemmatization was used to convert words into their stems, and to standardize vocabulary [14][45][9].

This dataset was fairly well balanced between the toxic and non-toxic classes, as well as the sarcastic and non-sarcastic classes [19][26]. But, there was a mild imbalance in sentiment polarity classes [42][45]. This was addressed by employing weighted loss functions instead of oversampling or undersampling while training the model [11][12][20]. This maintained the natural distribution of data and ensured that the minority classes played a role in learning [11][12]. In general, the results of the preprocessing process created a clean, semantically consistent and a reliable data set for feature extraction and classification process were carried out [14][16].

D. Feature Extraction and Interaction for Classical Models

In classical models, lexical and semantic properties were used to detect sarcasm and toxicity [7][32][33]. The unigrams, bigrams and trigrams were extracted and then TF-IDF was applied to the terms to emphasize the more informative ones and to suppress the effect of frequent terms [14][31]. Sentiment polarity features also included to reflect the emotional orientation and polarity changes in the text, which are sometimes an expression of sarcasm and irony [42][45][48][49]. Another set of incongruity measures were also added to identify semantic incongruity, such as positive sentiment paired with toxic expressions [32][33].

Sarcasm and toxicity in features was considered explicitly because it is common for seemingly benign features to form a combination that yields the toxic message [38][44]. These interactions were well captured by tree-based models like Random Forest and XGBoost with hierarchical splits [13, 39]. Other features designed were lexical diversity, punctuation, and syntactic features, which were useful for locating stylistic cues related to sarcasm and implicit toxicity [7][32][34][56]. These in combination offered a multi-dimensional view of text for better classification accuracy [7][36][38].

1) Transformer Models for Feature Representation:
Transformer Models for Feature Representation: DistilBERT and transformer based models were also employed for feature representation [12][41][50]. DistilBERT highlights the semantic meaning, syntactic relationships, and long-range dependencies between words, unlike TF-IDF [12][13][41]. This helps in identifying nuanced language forms like irony, sarcasm, and toxic in context [12][41][50].

E. Model Training

Logistic Regression, Random Forest, Support Vector Machine (SVM), XGBoost, and DistilBERT were used for the evaluation of the performance.The performance was evaluated using five models: Logistic Regression [11], Random Forest [12], Support Vector Machine (SVM) [13], XGBoost [39], and DistilBERT [41] [29]. TF-IDF vectors with engineered sentiment polarity and contextual incongruity features were employed as inputs for classical models (Logistic Regression, Random Forest, SVM, and XGBoost), which were able to capture lexical patterns and subtle semantic cues [7][14][32][6]. To avoid overfitting and to bring the best out of the algorithm, hyperparameter tuning was done using grid search with cross validation [13][39][41]. The number of trees, maximum depth, and learning rate were tuned, especially for tree based models, such as Random Forest and XGBoost [13][39][45].

DistilBERT was fine-tuned using pre-trained weights to take advantage of the contextual embeddings [12][41]. Early stopping on validation loss [12][14][33] and optimizer with learning rate scheduler (AdamW) [12][14] were adopted to avoid overfitting and to speed up the convergence, respectively. The data set was divided into training and testing data sets (80% and 20%) for assessing the generalization capability [11][12][19][15]. Weighted loss functions were applied to decrease the bias towards majority classes in the case of class imbalance [11][12][20]. In general, this training method worked well to capture sarcasm, toxicity, and context-dependent language patterns through feature engineering, hyperparameter optimization and thorough evaluation [7][12][36][38].

F. Evaluation Metrics

This study used Macro F1-Score as the main evaluation metric because it takes into account precision and recall of all classes [11][19][20]. This is especially crucial for imbalanced data sets where minority classes like sarcastic-toxic comments are not dominated by majority classes [11][19][28]. Particular emphasis was given to the F1-score of the sarcastic-toxic class because it is directly related to the model's ability to detect toxicity that is concealed in the sarcastic class, the primary purpose of this research [2][28][36].

Other assessment criteria also were applied to the detailed analysis. Precision and correctness of positive predictions were measured for each class [11][19] and recall showed the model's capacity to capture all relevant cases, showing the tradeoff between false positives and false negatives [11][19][20]. Analysing misclassification patterns, including misclassification of sarcastic-toxic comments as being neutral or purely toxic, was done using the pattern of confusion matrices [19][20][28]. Overall, such metrics fully evaluated the performance of the models in reproducing fine-grained, context-dependent language phenomena like sarcasm in toxic speech [7][12][36][38].

G. Exploratory Data Analysis

Exploratory data analysis (EDA) was performed to gain insight into the data characteristics and trend [7][30][36] before model development. Sarcasm's distribution was analyzed across the three platforms (TikTok, Instagram, and YouTube) to explore platform-specific differences, and heatmaps were employed to examine relationships between toxicity and sarcasm [7][30][36]. Sentiment distributions were examined on each platform to determine how many positive, negative and neutral comments were made, and correlation heatmaps were used to investigate the correlations between sarcasm, toxicity and sentiment [7][36][45].

Pairwise and scatter plots were also employed to examine the relationships between emotions in comments, sarcasm, and toxicity [7][36]. The EDA showed that sarcasm is more common in mixed-sentiment comments and moderately toxic comments [7][36][48]. They showed that features based on interactions and context-aware modeling were crucial for identifying subtle linguistic patterns, like sarcastic-toxic expressions [7][36][38]. In general, EDA helped in feature engineering and model selection by giving the user a clear insight of its dataset structure and complexity [7][36][38].

H. Experimental Setup

All experiments were done in Python with libraries for Machine Learning and Deep Learning [14][31][41]. The classical models including Logistic Regression, Random Forest and XGBoost were implemented using scikit-learn and XGBoost, whereas, DistilBERT model was implemented using the Hugging Face Transformers library which uses a PyTorch backend [12][14] [41] [25] [39]. All models had high-quality data inputs that were preprocessed using the Tokenization technique, TF-IDF vectorisation, removal of stopwords and normalization using the NLTK and Scikit-learn [14][31].

Additional pre-processing like padding, truncation, and generating attention masks were added for transformer models for DistilBERT [12][41][50]. Support for efficient fine-tuning was provided by running experiments on a GPU-equipped workstation. All datasets were split, using the same random split as the competitions, and the processing pipelines and feature engineering steps were the same for all models, to ensure fair comparison [11][12][14]. Hyperparameter tuning and cross validation were used throughout to improve performance and minimize bias [13][39][41][50]. Overall, this setup allowed for a reliable assessment of approaches for classical and transformer-based detection of sarcasm and toxic language of [1][2][6][12][36].

III. RESULTS

A. Descriptive Analysis

The data sources included social media posts from TikTok, Instagram, and YouTube about tariffs on America. Five key columns were mentioned: Platform, Comment, Toxicity, Sarcasm, and Sentiment. Data cleaning and data preprocessing were done after which sentiment distribution was analyzed. It was found that Neutral comments (93) were the most occurring ones signifying mostly factual or unemotional discussions, Negative comments (142) which signified criticism or disapproval of tariffs were the second most occurring, and Positive comments (58) which signified favourable or good discussions were the least occurring.

Correlations among the variables were evaluated by the following means:

1) Toxicity vs Sarcasm: A moderate correlation was found, meaning that while some people use sarcasm, they are not necessarily toxic and some sarcastic comments are toxic. This highlights that sarcasm is not an exclusive marker for toxicity in automated systems for detecting toxicity (Fig. 2).

2) Toxicity vs Sentiment: Toxicity was more common in negative comments, as offensive language was commonly found in criticism. The positive comments, however, had very little toxicity, indicating that positive opinions are less likely to be toxic. This indicates the relevance of sentiment in toxicity prediction (Fig. 2).

3) Sarcasm vs Sentiment:Sarcasm was found to be more prevalent in mixed or moderately negative comments rather than highly negative comments. That demonstrates that sarcasm is often humorous, ironic, or subtle in its criticism, but not in a rude or blatant way. The detection of sarcasm is easier if there is context available (Fig. 2), because it is strongly dependent on context.

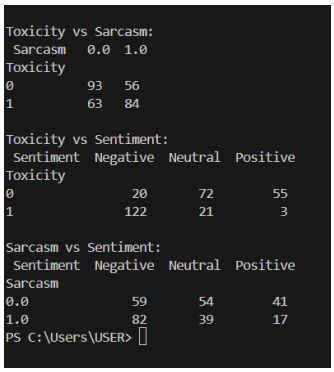


Figure 2. The Correlations among Variables

There was also an effect on sarcasm and toxicity with the platforms. TikTok had the most instances of sarcastic posts, indicating a younger user base that enjoys humor or irony in their short-form content. Instagram was next, and users had less sarcasm, depending on content type. YouTube had less sarcasm, but there was variation depending on content type. These differences suggest that generalizing sarcasm and toxicity across platforms is not possible without making adjustments, as the form in which they are expressed is influenced by the platform context and content format.

In general, toxic comments comprised 49.83% and sarcastic 47.30% of the data (Fig. 3). This suggests a very polarized discussion on tariffs, in which offensive language is heavily used. The sarcasm is also quite high, indicating that it is a typical indirect mode of criticism. What makes it difficult for automated systems to detect is that sarcasm can come across as toxic, while toxicity can be considered sarcasm. Taken together, these results indicate that the tariff debates are not only volatile and complex from an emotional point of view, but also from a linguistic perspective, where nuanced expressions of negativity are hard to be captured.

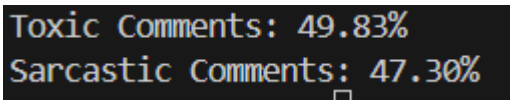


Figure 3. Percentages of Toxic and Sarcastic Comments

B. Model Evaluation

Macro F1-Score, F1-Score on the sarcastic-toxic subset, Accuracy, Precision, Recall and confusion matrices were used to compare the different models in both transformer-based and traditional machine learning. All of these metrics offer a holistic analysis of how well each model performs at detecting both toxicity and sarcasm, particularly when they are combined.

1) Transformer-Based Model (DistilBERT): DistilBERT outperformed the rest with a Macro F1-Score of 0.65 and an F1-Score of 0.67 on the sarcastic-toxic subset, making it very good at detecting toxic and sarcastic content. For toxicity detection, it had a good balance between accuracy, precision and recall, with values of 0.82, 0.86, and 0.70 respectively. For the sarcasm detection, it got an accuracy of 0.63, precision of 0.80 and recall of 0.39. The low recall suggests that the model may make a lot of correct predictions (high precision), but it fails to capture some sarcastic nuances. The confusion matrix validated that some sarcastic-toxic comments were classified as toxic or neutral which showed the challenge of sarcasm detection. Overall, DistilBERT was the most accurate model of capturing the toxicity in sarcasm.

2) Traditional Machine Learning Models: Logistic Regression had the worst performance with a Macro F1-Score of 0.60 and an F1-Score of 0.55 on the sarcastic-toxic subset. It was moderately balanced, but it had a lot of problems with sarcasm detection, misclassifying sarcastic-toxic comments. Random Forest had a slightly better performance with Macro F1-Score of 0.62 and F1-Score of 0.56, due to ensemble-based feature interactions but limited sarcasm recall. SVM model performed slightly better in detecting sarcastic-toxic words with the F1-Score of 0.57 and Macro F1-Score of 0.62, although still not well in the context-dependent sarcasm. The best classical model was XGBoost which recorded a Macro F1-Score of 0.65 and an F1-Score of 0.65, very close to DistilBERT. It performed better than other classical models in toxicity and sarcasm detection due to its capability in modeling non-linear feature interactions, and less misclassifications of sarcastic-toxic comments in its confusion matrix.

C. Comparative Analysis

Finally, Best Model: DistilBERT performed better than all the classical machine learning models, particularly when detecting sarcastic-toxic utterances. It is designed in a transformer architecture that can capture the rich contextual dependency, which helps detect subtle linguistic patterns that are hard for traditional models to capture. Hence, DistilBERT proves to be the most appropriate model to accurately identify sarcasm-masked toxicity in social media text. Amongst classical models, Best Classical Model: XGBoost performed best and results were close to DistilBERT. Its gradient boosting mechanism is able to model the complex non-linear relationships in the feature space, which makes it a good modeler for toxicity and sarcasm detection. It's also a more efficient alternative to transformer-based models.

Moderate performing models were the Baseline Models: Logistic Regression and Random Forest. These baselines outperformed SVM slightly but still failed to achieve the best performance as compared to XGBoost and DistilBERT, especially in the task of detecting sarcasm-toxicity. The results illustrate the problems of traditional models with respect to context-dependent language.

The most challenging to classify were the sarcastic-toxic comments in all models. The existence of sarcasm examples often led to misclassifications, suggesting the need for context dependent models. In general, transformer-based DistilBERT achieved the highest performance, whereas XGBoost demonstrated the best classical solution, highlighting the significance of context-aware methods for identifying subtle and nuanced toxic content in social media.

IV. DISCUSSION

The results show the challenges faced in identifying toxicity in online conversations, while also demonstrating the relative strengths of various modeling methods. DistilBERT outperformed all other models in terms of all the metrics which demonstrates its superior ability in learning contextual dependencies, irony and minor changes in sentiment. It outperforms classical models that focus more on surface features in their use of semantic representations. Traditional models performed best using the sentiment and incongruity features alone, and in some cases, were close to the DistilBERT scores, demonstrating that gradient boosting can capture complex feature interactions when added to sentiment and incongruity features. But it is not as good as transformer-based models for subtle and indirect sarcasm.

In the results of the platform analysis, it was determined that the amount of sarcastic-toxic comments is the highest on the TikTok platform, followed by Instagram, YouTube platform, and Facebook platform. It was determined that the culture of the platforms affects the expression of sarcasm and toxicity. Further evidence is provided by the fact that the higher prevalence in mixed-sentiment contexts further supports the need for context-aware detection models.

The results indicate that transformer models such as DistilBERT could be used to enhance automated text moderation systems to identify more specific types of toxic content, whereas XGBoost can serve as an effective computationally efficient solution in resource-constrained environments. The dataset's scope of American tariff discussions means that it may be hard to generalize to other contexts, and may include biases from human annotation of sarcasm. This study differs from previous work by examining the relationships between sarcasm and toxicity together, demonstrating that many times, there is a malicious intent behind sarcasm and that the context is crucial to their detection.

V. CONCLUSION AND FUTURE WORK

This article was a systematic comparison of transformer and classic machine learning models at the task of identifying sarcasm-covered toxicity in online remarks about American tariffs. The findings show that DistilBERT is significantly better than classical models, and context-sensitive, pre-trained language models are important to capture delicate linguistic information, e.g., sarcasm and implicit toxicity. XGBoost performed competitively among classical methods, indicating that the well-tuned ensemble models can also be used as alternatives to computationally demanding models in case of limited resources.

The platform-specific analysis showed that sarcastic-toxic posts are most likely to be found on Tik Tok, then on Instagram and YouTube, which showed that the content of social media and the behavior of users contribute heavily to the articulation of sarcasm and toxicity. The results highlight the shortcomings of polarity models or superficial feature models and the need to include contextual details to ensure robust automated moderation.

The weaknesses of this work are that the dataset is only limited to one topical area, the use of human annotations to detect sarcasm and toxicity, and the production does not include other multimodal elements that might contain other sarcastic indicators like pictures or emojis. The future work can expand on this study by looking at cross-domain datasets, combining multimodal data, and testing more recent transformer models like GPT-based models in detecting sarcasm-masked toxicity better. Besides, explainability and interpretability practices may be implemented to

achieve transparency in automated moderation decisions, as is the case with recent works on transformer-based sarcasm and toxicity recognition [2], [8], [20], [41], [50]. In general, this research will add to the comprehension of the issues of sarcasm making the detection of toxicity more challenging and shows that transformer-based models, especially DistilBERT, can be a powerful tool to control the intricate online communication.

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GGI Gamanayake is a final year BSc Applied Data Science Communication undergraduate student at General Sir John Kotelawala Defence University, Sri Lanka. Her field of study is natural language processing, machine learning and social media analytics. She has participated in research related to the detection of sarcasm-masked toxicity using transformer-based models, specifically in the field of NLP based text classification, and model evaluation



MGKS Gamage, a final year undergraduate at Sri Lanka's General Sir John Kotelawala Defence University, in the same programme. She's interested in remote sensing, geospatial data analysis, and environmental data science. She has participated in research projects that include analysis of structured datasets in the environmental , geospatial areas and statistical analysis.



DD Weerasuriya is a student from the General Sir John Kotelawala Defence University, Sri Lanka in the Applied Data Science Communication BSc undergraduate program at the end of her fourth year of study. She has academic interests in data visualization, exploratory data analysis and business intelligence. She has worked with data sets to explore and visualize data in Power BI, and with statistical analysis methods.